

MODERN INSTITUTE OF TECHNOLOGY AND RESEARCH CENTRE, ALWAR

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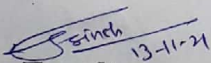
UNIT: I

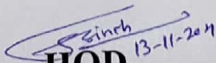
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References:

1. R. K. Rajput, Basic Mechanical Engineering, Laxmi Publications
2. B. Agrawal, Mechanical Engineering, John Wiley & Sons


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LECTURE -01 (UNIT-01)**Syllabus :-Introduction to Thermodynamics and Terminology****CONCEPT OF THERMAL ENGINEERING: -**

- ❖ Thermal engineering deals with the concepts of thermodynamics related to heat, temperature, pressure, work etc.
- ❖ It describes the various work producing & work absorbing devices, their performance and their efficiency.

Various concepts of thermal engineering are discussed below.

2. 1 THERMODYNAMIC SYSTEMS

- ❖ A fixed area or region or space in which different thermodynamic processes take place is called 'System'.

Thermodynamic system:

Thermodynamic system can be defined as a definite area or space where some thermodynamic process takes place. It is the region where transformation of energy or mass to be studied.

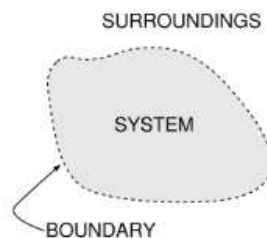
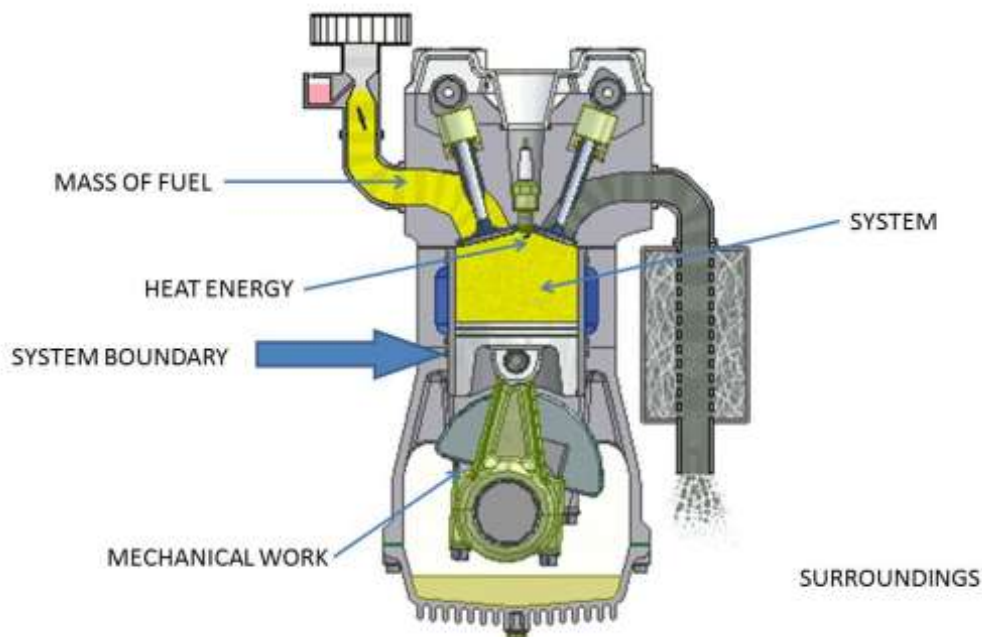


Fig 1.1 Thermodynamic system

Various terms associated with thermodynamic system:

- i. **System:** A definite amount of liquid or gas contained in boundary is known as system.
- ii. **Boundary:** The layer enclosing the system is known as boundary. The boundary may be flexible or rigid.
- iii. **Surroundings:** All the matter outside the boundary is called surroundings.
- iv. **Working Fluid:** The working fluid is a liquid or gas that absorbs or transmits energy in a thermodynamic system.

IC ENGINE (SYSTEM)



TERMINOLOGY:-

State, Path, Process and Cycle.

State of a system:

The state of the system is the condition of the system at any particular moment which can be identified by its properties such as pressure, volume, temperature etc.

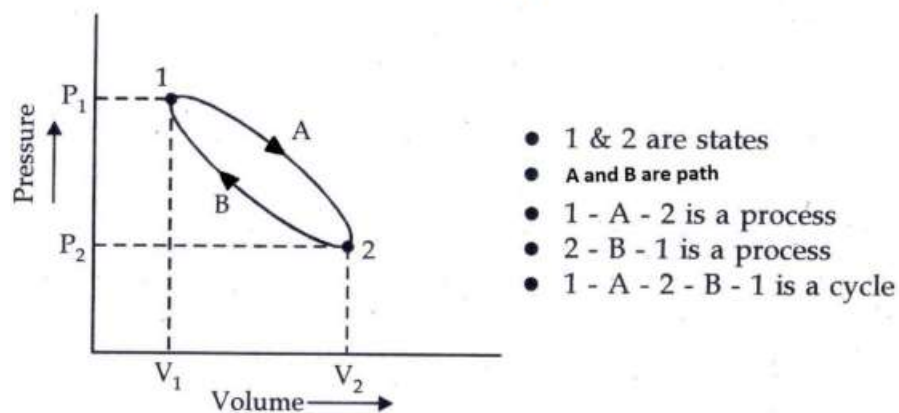


Fig 1.5 State, Path, Process and cycle

From fig 1.2 point '1' represents initial state of a system with properties P_1 , V_1 and T_1 . Due to energy transfer, the new state is represented by another point '2' with properties P_2 , V_2 and T_2 .

Path of a System:

It is the path through which a system passes from an initial equilibrium state to a final equilibrium state.

In fig. 1.5 **A** and **B** are path.

Thermodynamic Process:

When a system changes its state from one equilibrium state to another equilibrium state, then the path of successive state through which a system has passed is known as process or thermodynamic Process.

In fig.1.5 **1-A-2** and **2-B-1** are process.

Thermodynamic Cycle:

When a process is performed on a system in such a way that the final state is identical with initial state, it is known as cycle or thermodynamic cycle.

In fig 1.5 **1-A-2-B-1** is a cycle.

Define quasi static process: -

Q. Define Quasi-static process.

Quasi-Static Process

When the process is carried out in such a rate that at every intermediate points, the equilibrium conditions can be described and fixed, then the process is known as Quasi-static process.

Each state in Quasi-static process may be considered as an equilibrium state. Quasi-static process is also called as reversible process. The process can be reversed and the system reacts with same equilibrium states.

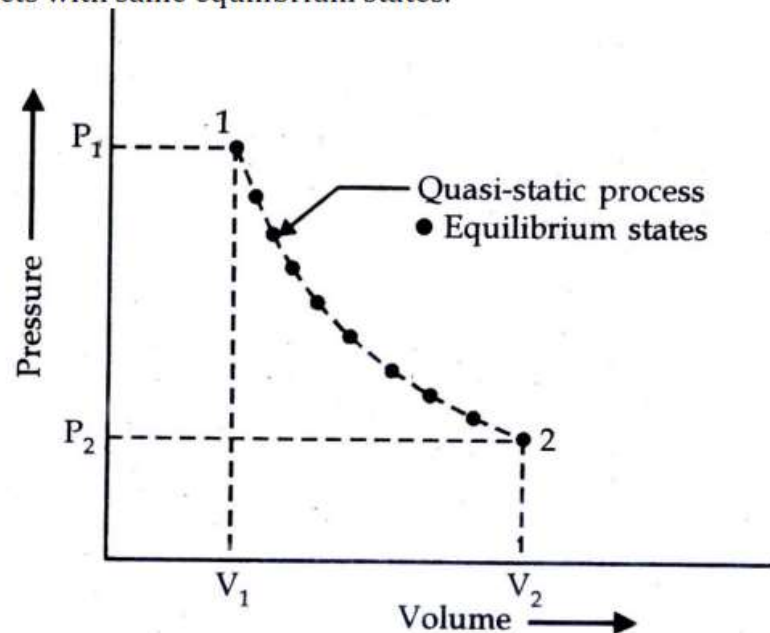


Fig 1.8 Quasi-Static Process

REFERENCES:-

- 1. R. K. Rajput, Basic Mechanical Engineering, Laxmi Publications**
- 2. B. Agrawal, Mechanical Engineering, John Wiley & Sons**



LECTURE -02 (UNIT-01)**Syllabus :-Thermodynamic properties, closed and open systems****Thermodynamic systems:**

The Thermodynamic system can be classified into following:

- a) Closed system
- b) Open system
- c) Isolated system

a) Closed system: In a closed system heat and work crosses the system boundary but the mass remains constant.

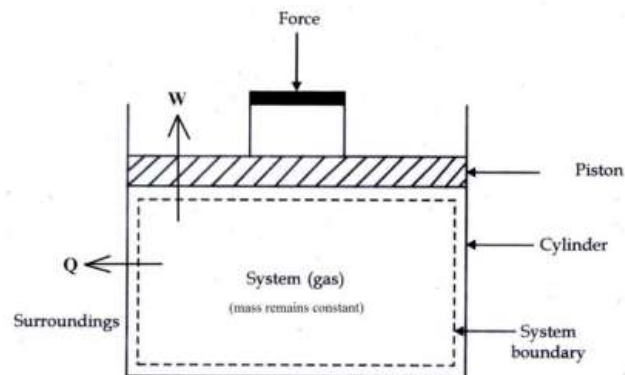
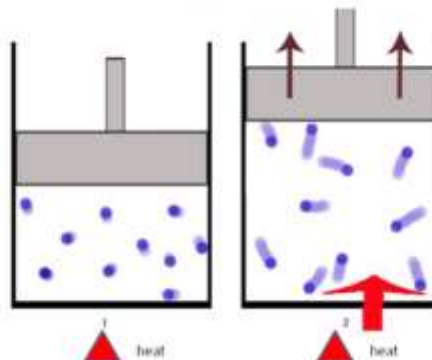


Fig. 1.2 Closed System

Ex.: Consider a system of gas in a cylinder with piston arrangement as shown in fig. When the gas is heated from the external source, the temperature and volume of the gas increases and the piston will move up and the boundary of the system also moves causes the heat and work crosses the boundary but mass remains fixed.

Hence it is a system of fixed mass and identity whose boundary are determined by the space occupied by the working substance.

a) Closed system:
Heat and work crosses the system boundary but the mass remains constant.



b) Open system: In this system, the mass of the working substance crosses the system boundary. The heat and work may also cross the system boundary.

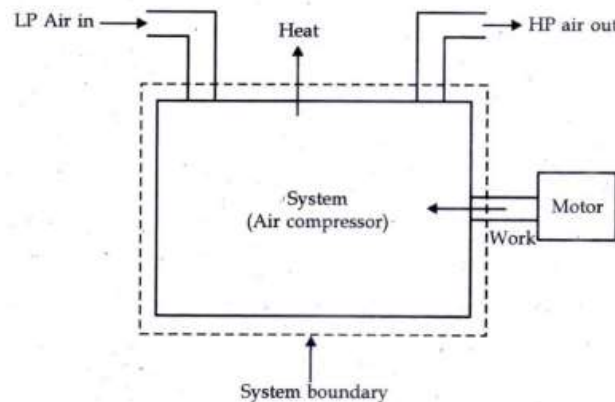
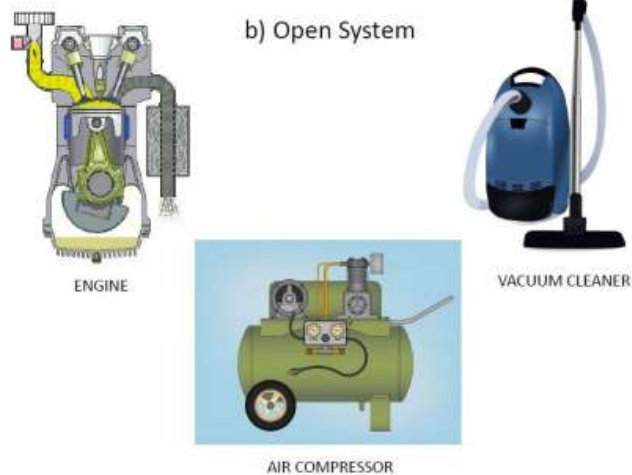


Fig 1.3 Open system

Ex.: The operation of air compressor i.e. the working substance in air compressor crosses the boundary; low pressure air enters the compressor and leaves the high pressure air. The work crosses the boundary through the driving shaft and heat also crosses the boundary through cylinder walls.



c) Isolated system: In this system mass is fixed and no heat and work crosses the system boundary.

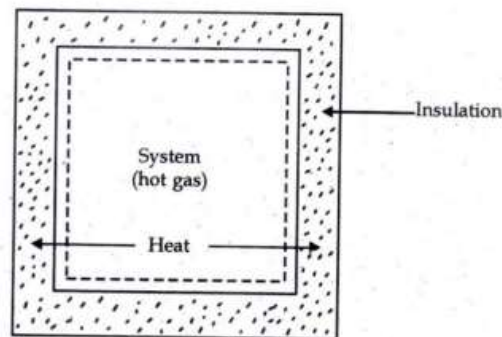


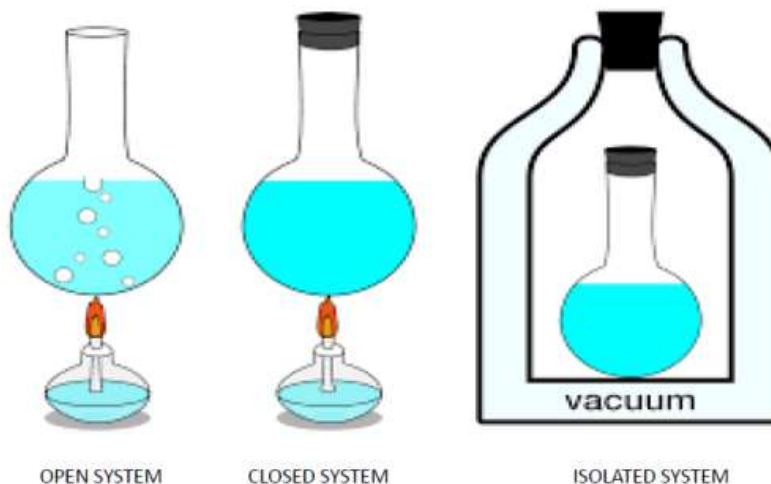
Fig 1.4 Isolated system

Ex.: A hot gas contained in an insulated container. The isolated system does not permit the mass and energy (heat energy and work energy) to cross the system boundary.

Q. State the comparison between closed system and open system.

Comparison between closed system and open system

CLOSED SYSTEM	OPEN SYSTEM
1] It does not permit mass to cross the system boundary but, heat and work crosses the system boundary	1] It permits mass, heat and work to cross the system boundary
2] Mass is fixed	2] Mass is not fixed
3] Ex. Gas heated in closed piston and cylinder	3] Ex. Air compressor



Properties of a system: -

Properties of a System:

Thermodynamic properties can be classified into following

- 1] Extensive property.
- 2] Intensive property.

1] Extensive properties:

The properties which depend upon the mass of the system are called extensive properties.

Ex:- Total volume, total mass, total energy, enthalpy, entropy, internal energy etc.

In extensive property, the quantity of matter in a system is divided into a number of parts then the magnitude of properties is equal to the sum of their values for individual parts of the system.

2] Intensive properties:

The properties which do not depend upon the mass of the system are called intensive properties.

Ex:- Pressure, temperature, density, specific heat, specific volume etc. The values of these properties are obtained per unit mass.

In intensive properties, if the quantity of a matter in a system is divided into a number of parts, then the magnitude of properties of a system is not equal to the sum of their values for individual parts of the system.

Q. Differentiate between extensive and intensive properties of a system. Give three examples for each.

Differentiate between extensive and intensive properties of a system

EXTENSIVE PROPERTIES	INTENSIVE PROPERTIES
1] Depend upon mass	1] Do not depend upon mass
2] These properties are obtained on total mass	2] These properties are obtained per unit mass
3] The magnitude of system is equal to sum of their individual parts of a system	3] The magnitude of system is not equal to sum of their individual parts of a system
4] Ex: Total mass, Total Energy, Total Volume, enthalpy etc.	4] Ex: Temperature, Pressure, Density, specific heat etc.

Q. State Zeroth law, First law and Second law of thermodynamics.

Laws of thermodynamics:

1] Zeroth law of Thermodynamics: "When two systems are thermal equilibrium with a third system, then the two systems are also in thermal equilibrium with one another."

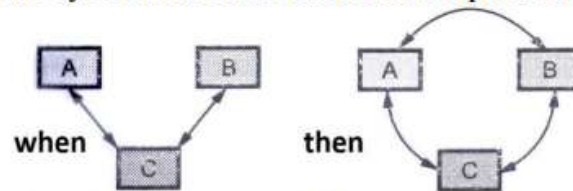


Fig 1.9 **Zeroth law of Thermodynamics**

2] First law of Thermodynamics: It deals with law of conservation of energy, it has two statements:

a) "The heat and mechanical work are mutually convertible."

$$\oint \delta Q = \oint \delta W$$

b) "The energy can neither be created nor destroyed though it can be transformed from one form to another."

$$Q = dU + W$$

$$\text{i.e } Q = (U_2 - U_1) + W$$

Where, Q = Heat supplied
 $dU = (U_2 - U_1)$ = Change in internal energy
 W = Work Done

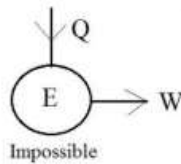
[Note: Heat supplied (Q) is used in increasing the internal energy from U_1 to U_2 of a gas and doing some external work (W). Thus conservation of energy]

Q. State two statements of second law of thermodynamics

3] Second law of Thermodynamics:

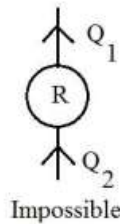
a) Kelvin-Plank statement: "It is impossible to construct an engine working on a cyclic process, whose sole purpose is to convert all the heat energy supplied to it into equivalent amount of work"

Ex: Heat Engine



b) Clausius statement: "It is impossible for a self-acting machine, working in a cyclic process, to transfer heat from cold body to hot body without the aid of an external agency"

Ex: Refrigerator



REFERENCES:-

1. R. K. Rajput, Basic Mechanical Engineering, Laxmi Publications
2. B. Agrawal, Mechanical Engineering, John Wiley & Sons

ELEMENT OF MECHANICAL ENGINEERING**LECTURE -03 (UNIT-01)****Syllabus :-Flow and Non-Flow Processes, Gas Laws****FLOW PROCESSES**

The processes which happen in an open system are called flow processes. As already studied, an open thermodynamic system is one in which there is not a fixed quantity or mass of gas/vapours but the working substance continuously flow in or out of the system. When this flowing substance or fluid exchange heat & work energy with the surrounding, the process is called flow process. Examples of this type of system are IC engines, centrifugal pump, steam turbines, water turbines etc.

These flow systems may further be of the two types as **steady flow system** and **unsteady flow system**.

1. Steady flow systems

The system through which the mass flow rate is constant i.e.

$$\text{Mass Input} = \text{Mass Output}$$

is a steady flow system. The state of working substance at any point in this system remains constant. Examples: Most actual thermodynamic equipment work as steady flow system under steady state conditions. Examples are IC engine, Compressors, steam turbines etc.

2. Un-steady flow system

The open system through which the mass flow rate of working substance is not constant is un-steady flow system. Here the study of unsteady flow system is beyond the scope of syllabus.

Non-Flow Processes

The boundary of a non-flow process can be fixed, moving or imaginary.

These are compression and expansion processes on gases in a cylinder with complete leak proof. In these there is only energy transfer with zero mass transfer.

These non flow processes can be the followings:

(i) constant pressure process

While heating at constant pressure

$$dU = \delta q - p dv$$

In an isobaric compression, heat is stored in the form of enthalpy.

(note the use of δ and d in the equation).

δ is used with heat supplied or rejected

d for change in a quantity such as volume or internal energy

(ii) constant volume process

$$\delta q = dU \text{ since } \delta W = 0$$

During a iso-choric (Constant volume) process, work done is zero and the total energy changes into internal energy.

(iii) constant temperature process

$$\delta q = \delta W \text{ since } dU = 0.$$

(iv) reversible adiabatic process

During a reversible adiabatic process, there is no friction.

No heat is supplied or rejected. Therefore $\delta W = dU$ since $\delta Q = 0$

A process with no heat gain or heat loss is an adiabatic process.

(v) poly-tropic process

During a poly-tropic process, $\delta Q = \delta W$ –work lost in friction

$p V^n = C$ is a mathematical form of a poly-tropic process in which no parameter is constant.

(vi) constant internal energy process

When gravity, magnetic, electrical, motion and capillary effects are negligible. Then the total energy (E) is equal to the internal energy (U).

Gas Laws: Overview

Created in the early 17th century, the gas laws have been around to assist scientists in finding volumes, amount, pressures and temperature when coming to matters of gas. The gas laws consist of three primary laws: Charles' Law, Boyle's Law and Avogadro's Law (all of which will later combine into the General Gas Equation and Ideal Gas Law).

Introduction

The three fundamental gas laws discover the relationship of pressure, temperature, volume and amount of gas. Boyle's Law tells us that the volume of gas increases as the pressure decreases. Charles' Law tells us that the volume of gas increases as the temperature increases. And Avogadro's Law tell us that the volume of gas increases as the amount of gas increases. The ideal gas law is the combination of the three simple gas laws.

Ideal Gases

Ideal gas, or perfect gas, is the theoretical substance that helps establish the relationship of four gas variables, pressure (P), *volume(V)*, the *amount of gas(n)* and *temperature(T)*. It has characters described as follow:

1. The particles in the gas are extremely small, so the gas does not occupy any spaces.
2. The ideal gas has constant, random and straight-line motion.
3. No forces between the particles of the gas. Particles only collide elastically with each other and with the walls of container.
4. No forces between the particles of the gas. Particles only collide elastically with each other and with the walls of container.

Real Gases

Real gas, in contrast, has real volume and the collision of the particles is not elastic, because there are attractive forces between particles. As a result, the volume of real gas is much larger

than of the ideal gas, and the pressure of real gas is lower than of ideal gas. All real gases tend to perform ideal gas behavior at low pressure and relatively high temperature.

The **compressibility factor (Z)** tells us how much the real gases differ from ideal gas behavior.

$$Z = PV / nRT \quad (\text{Gas Laws.1})$$

For ideal gases, $Z=1$ & for real gases, $Z \neq 1$

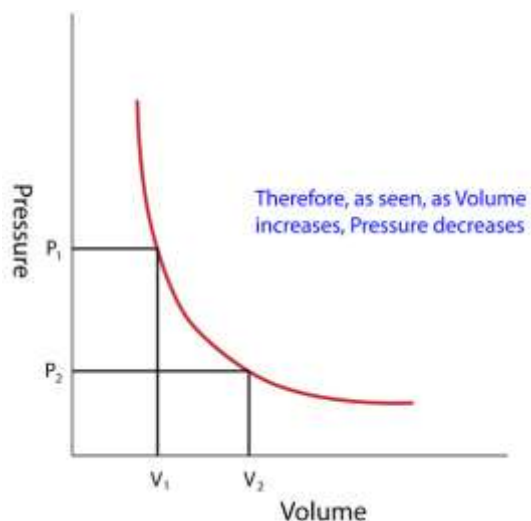
Boyle's Law

In 1662, Robert Boyle discovered the correlation between *Pressure (P)* and *Volume (V)* (assuming *Temperature (T)* and *Amount of Gas (n)* remain constant):

$$P \propto (1/V) \rightarrow PV = x \quad (\text{Gas Laws.2})$$

where x is a constant depending on amount of gas at a given temperature.

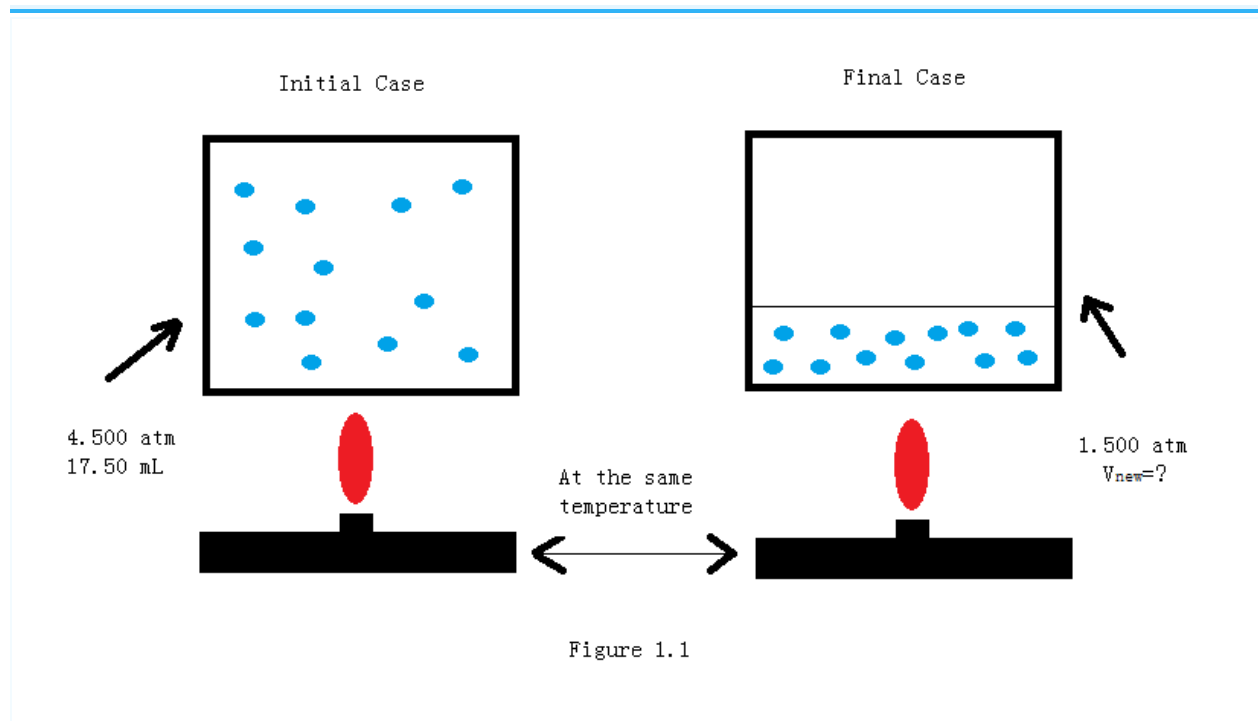
- Pressure is inversely proportional to Volume



- Another form of the equation (assuming there are 2 sets of conditions, and setting both constants to each other) that might help solve problems is:

$$P_1 V_1 = x = P_2 V_2 \quad (\text{Gas Laws.3})$$

Example 1.1 A 17.50 mL sample of gas is at 4.500 atm. What will be the volume if the pressure becomes 1.500 atm, with a fixed amount of gas and temperature?



Solution

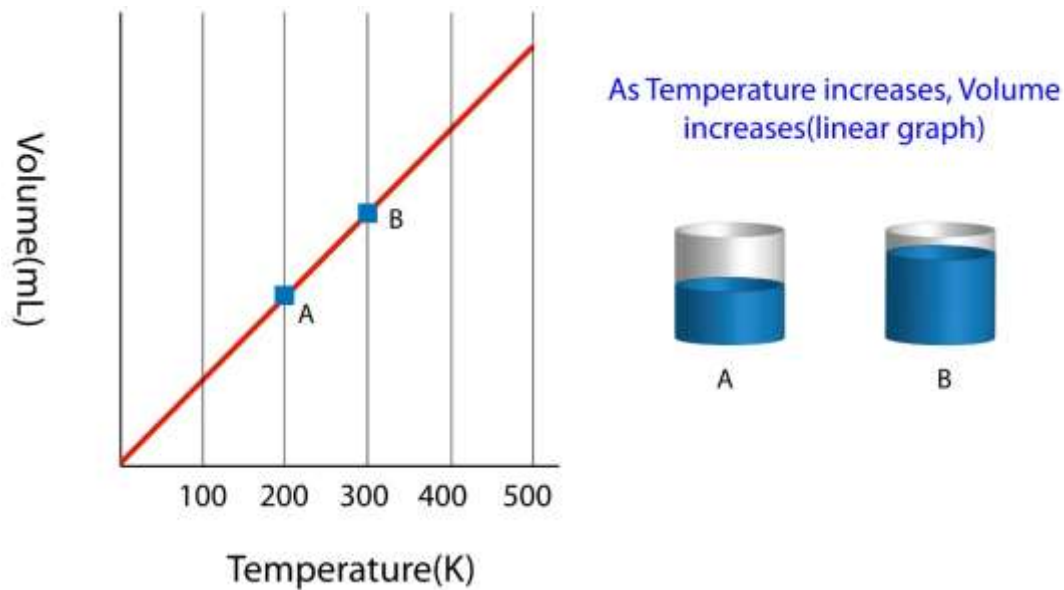
$$\begin{aligned}
 V_2 &= (P_1 \cdot V_1) / P_2 \\
 &= (4.500 \text{ atm} \cdot 17.50 \text{ mL}) / (1.500 \text{ atm}) \\
 &= 52.50 \text{ mL}
 \end{aligned}$$

Charles' Law

In 1787, French physicists Jacques Charles, discovered the correlation between *Temperature* (T) and *Volume* (V) (assuming *Pressure* (P) and *Amount of Gas* (n) remain constant):

$$V \propto T \rightarrow V = yT \quad (\text{Gas Laws.7})$$

where y is a constant depending on amount of gas and pressure. Volume is directly proportional to Temperature



Another form of the equation (assuming there are 2 sets of conditions, and setting both constants to each other) that might help solve problems is:

$$V_1 T_1 = V_2 T_2$$

(Gas Laws.8)

Example 1.2

A sample of Carbon dioxide in a pump has volume of 20.5 mL and it is at 40.0 °C. When the amount of gas and pressure remain constant, find the new volume of Carbon dioxide in the pump if temperature is increased to 65.0 °C.

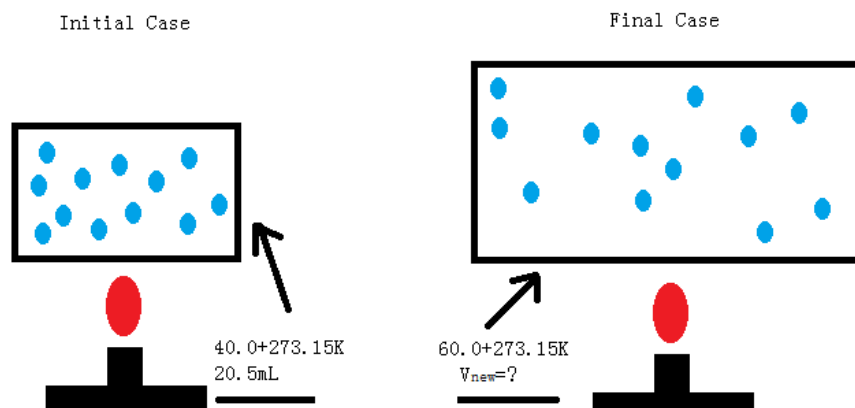


Figure 1.2

SOLUTION

$$\begin{aligned}
 V_2 &= (V_1 \cdot T_2) / (T_1) && \text{(Gas Laws.9)} \\
 &= 20.5 \text{ mL} \cdot (60 + 273.15 \text{ K}) / (40 + 273.15 \text{ K}) \\
 &= 22.1 \text{ mL}
 \end{aligned}$$

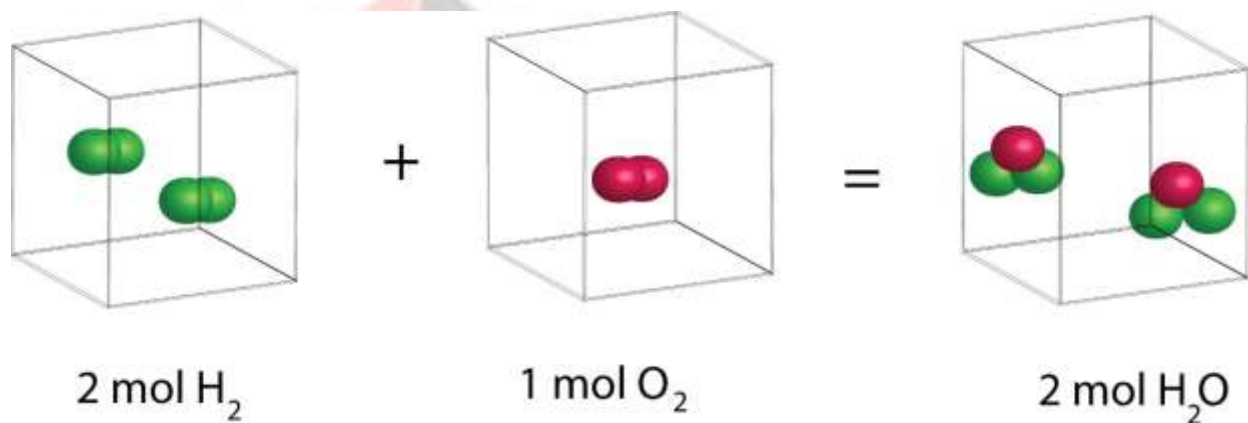
Avogadro's Law

In 1811, Amedeo Avogadro fixed Gay-Lussac's issue in finding the correlation between the *Amount of gas*(n) and *Volume*(V) (assuming *Temperature*(T) and *Pressure*(P) remain constant):

$$V \propto n \rightarrow V = zn \quad \text{(Gas Laws.12)}$$

where z is a constant depending on Pressure and Temperature.

- Volume(V) is directly proportional to the Amount of gas(n)



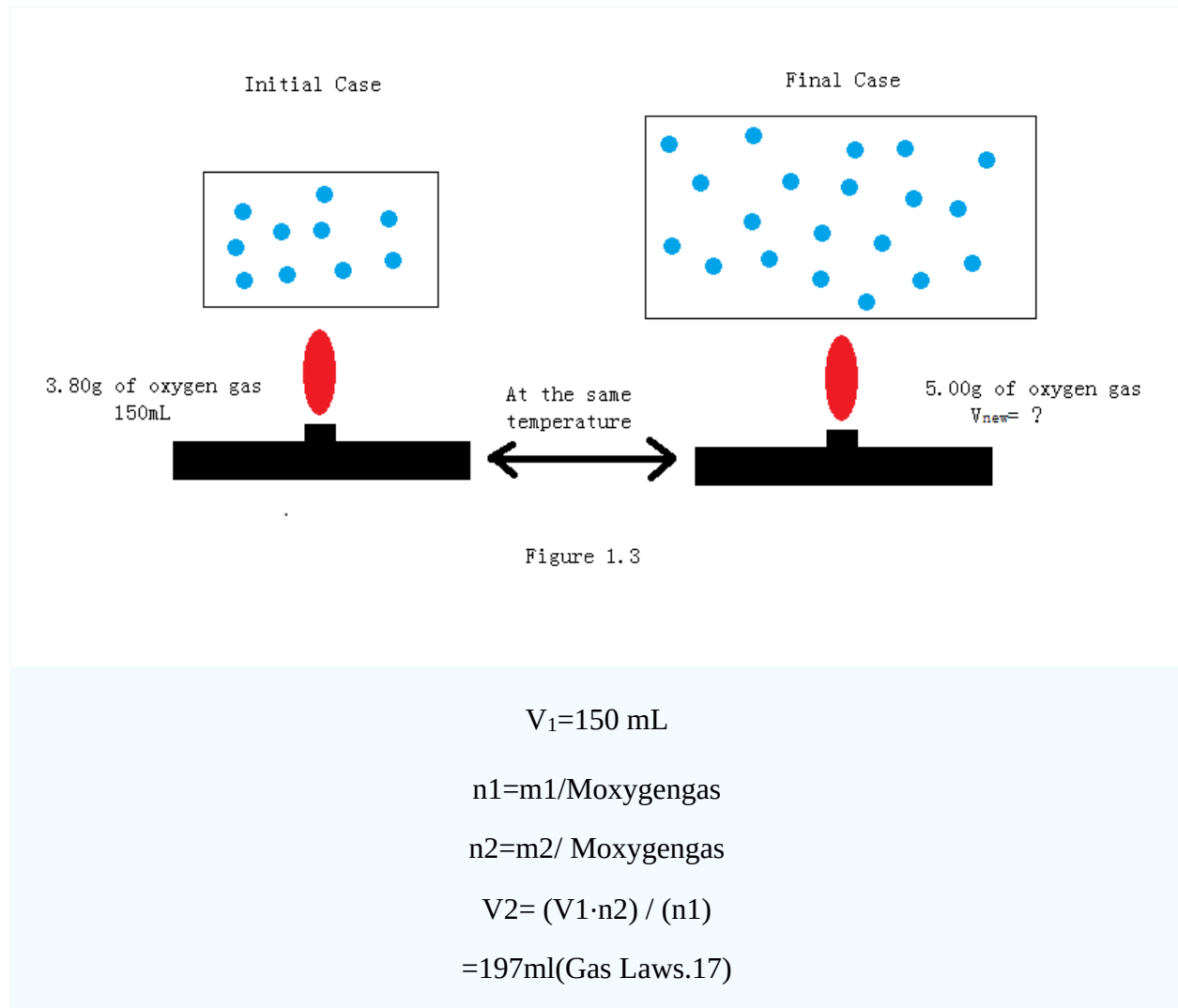
Another form of the equation (assuming there are 2 sets of conditions, and setting both constants to each other) that might help solve problems is:

$$(P_1 / n_1) = z = (P_2 / n_2) \quad \text{(Gas Laws.13)}$$

Example 1.3

A 3.80 g of oxygen gas in a pump has volume of 150 mL. constant temperature and pressure. If 1.20g of oxygen gas is added into the pump. What will be the new volume of oxygen gas in the pump if temperature and pressure held constant?

Solution



Ideal Gas Law

The ideal gas law is the combination of the three simple gas laws. By setting all three laws directly or inversely proportional to Volume, you get:

$$V \propto (nT)/P \quad (\text{Gas Laws.18})$$

Next replacing the directly proportional to sign with a constant(R) you get:

$$V = (RnT) / P \quad (\text{Gas Laws.19})$$

And finally get the equation:

$$PV=nRT \quad \text{(Gas Laws.20)}$$

where P= the absolute pressure of ideal gas

- V= the volume of ideal gas
- n = the amount of gas
- T = the absolute temperature
- R = the gas constant

Here, R is called the gas constant. The value of R is determined by experimental results. Its numerical value changes with units.

$$\begin{aligned} R &= \text{gas constant} = 8.3145 \text{ Joules} \cdot \text{mol}^{-1} \cdot \text{K}^{-1} \text{ (SI Unit)} \\ &= 0.082057 \text{ L} \cdot \text{atm} \cdot \text{K}^{-1} \cdot \text{mol}^{-1} \end{aligned}$$

REFERENCES:-

1. R. K. Rajput, Basic Mechanical Engineering, Laxmi Publications
2. B. Agrawal, Mechanical Engineering, John Wiley & Sons

ELEMENT OF MECHANICAL ENGINEERING**LECTURE -04 (UNIT-01)****Syllabus :-** Laws of Thermodynamics, Internal Energy**4.1 ZEROTH LAW OF THERMODYNAMICS**

‘Zeroth law of thermodynamics’ states that if two systems are each equal in temperature to a third, they are equal in temperature to each other.

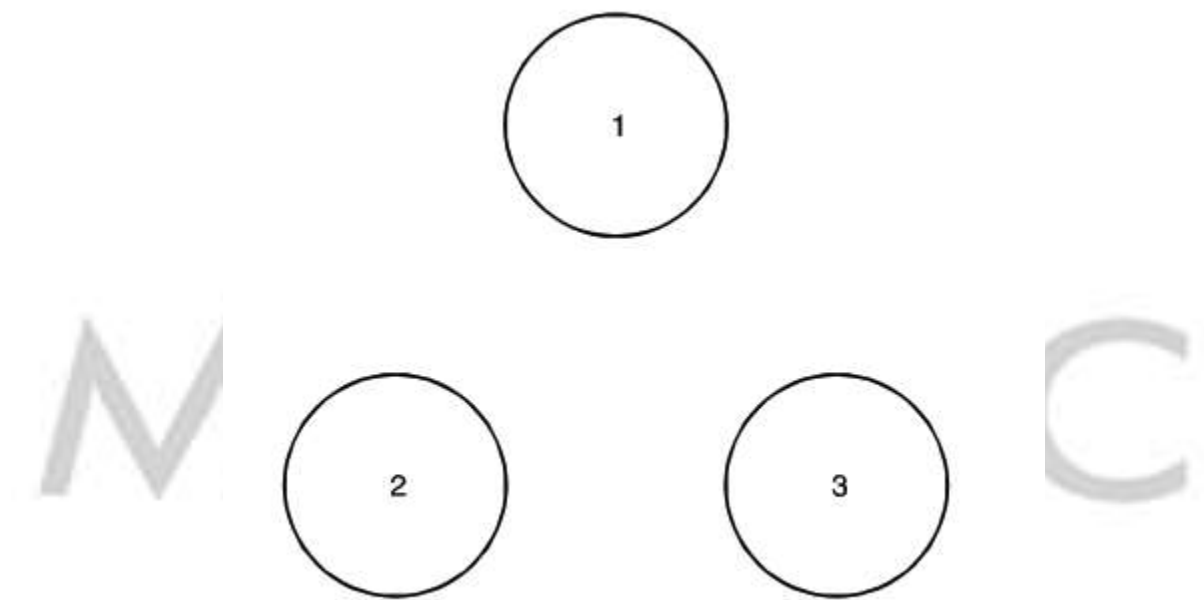


Fig. 2.7. Zeroth law of thermodynamics.

Example. Refer Fig. 2.7. System ‘1’ may consist of a mass of gas enclosed in a rigid vessel fitted with a pressure gauge. If there is no change of pressure when this system is brought into contact with system ‘2’ a block of iron, then the two systems are equal in temperature (assuming that the systems 1 and 2 do not react each other chemically or electrically). Experiment reveals that if system ‘1’ is brought into contact with a third system ‘3’ again with no change of properties then systems ‘2’ and ‘3’ will show no change in their properties when brought into contact provided, they do not react with each other chemically or electrically. Therefore, ‘2’ and ‘3’ must be in equilibrium.

This law was enunciated by R.H. Fowler in the year 1931. However, since the first and second laws already existed at that time, it was designated as zeroth law so that it precedes the first and second laws to form a logical sequence.

4.2 INTERNAL ENERGY

It is the heat energy stored in a gas. If a certain amount of heat is supplied to a gas the result is that temperature of gas may increase or volume of gas may increase thereby doing some external work or both temperature and volume may increase; but it will be decided by the conditions under which the gas is supplied heat. If during heating of the gas the temperature increases its internal energy will also increase.

Joule's law of internal energy states that internal energy of a perfect gas is a function of temperature only. In other words, internal energy of a gas is dependent on the temperature change only and is not affected by the change in pressure and volume.

We do not know how to find the absolute quantity of internal energy in any substance; however, what is needed in engineering is the change of internal energy (ΔU).

4.3 LAW OF CONSERVATION OF ENERGY

In the early part of nineteenth century the scientists developed the concept of energy and hypothesis that it can be neither created nor destroyed; this came to be known as the law of the conservation of energy.

4.3.1 FIRST LAW OF THERMODYNAMICS

“When a system undergoes a thermodynamic cycle then the net heat supplied to the system from the surroundings is equal to net work done by the system on its surroundings.

or
$$\oint dQ = \oint dW$$

where $\oint$ represents the sum for a complete cycle.

— The First Law of Thermodynamics may also be stated as follows :

“Heat and work are mutually convertible but since energy can neither be created nor destroyed, the total energy associated with an energy conversion remains constant”.

Or

— **“No machine can produce energy without corresponding expenditure of energy, i.e., it is impossible to construct a perpetual motion machine of first kind”.**

Fig. 4.1 shows the experiment for checking first law of thermodynamics.

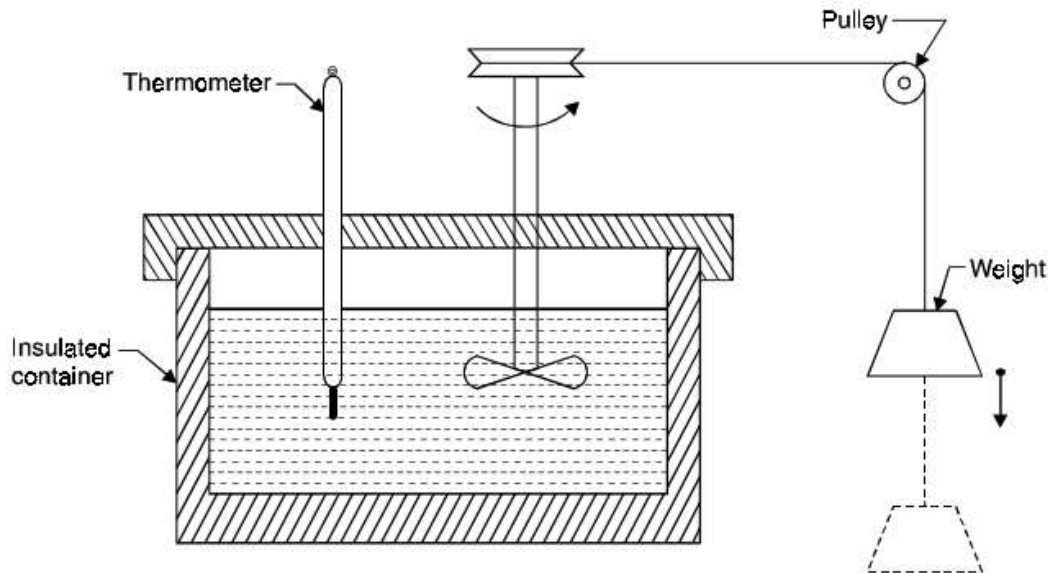


Fig. 4.1. Heat and work.

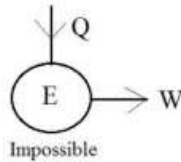
The work input to the paddle wheel is measured by the fall of weight, while the corresponding temperature rise of liquid in the insulated container is measured by the thermometer. It is already known to us from experiments on heat transfer that temperature rise can also be produced by heat transfer.

Q. State two statements of second law of thermodynamics

3] Second law of Thermodynamics:

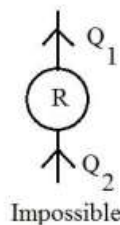
a) Kelvin-Plank statement: "It is impossible to construct an engine working on a cyclic process, whose sole purpose is to convert all the heat energy supplied to it into equivalent amount of work"

Ex: Heat Engine



b) Clausius statement: "It is impossible for a self-acting machine, working in a cyclic process, to transfer heat from cold body to hot body without the aid of an external agency"

Ex: Refrigerator



Joule's Law

Joule's law states as follows :

"The internal energy of a perfect gas is a function of the absolute temperature only."

i.e., $u = f(T)$

REFERENCES:-

1. R. K. Rajput, Basic Mechanical Engineering, Laxmi Publications
2. B. Agrawal, Mechanical Engineering, John Wiley & Sons

ELEMENT OF MECHANICAL ENGINEERING

LECTURE -05 (UNIT-01)

Syllabus :- Application of First Law in Heating and Expansion of Gases in Non-Flow Processes only

APPLICATION OF FIRST LAW OF THERMODYNAMICS TO NON-FLOW OR CLOSED SYSTEM

1. Reversible Constant Volume (or Isochoric) Process ($v = \text{constant}$):

In a constant volume process the working substance is contained in a rigid vessel, hence the boundaries of the system are immovable and no work can be done on or by the system, other than paddle-wheel work input. It will be assumed that 'constant volume' implies zero work unless stated otherwise.

Fig. 4.5 shows the system and states before and after the heat addition at constant volume.

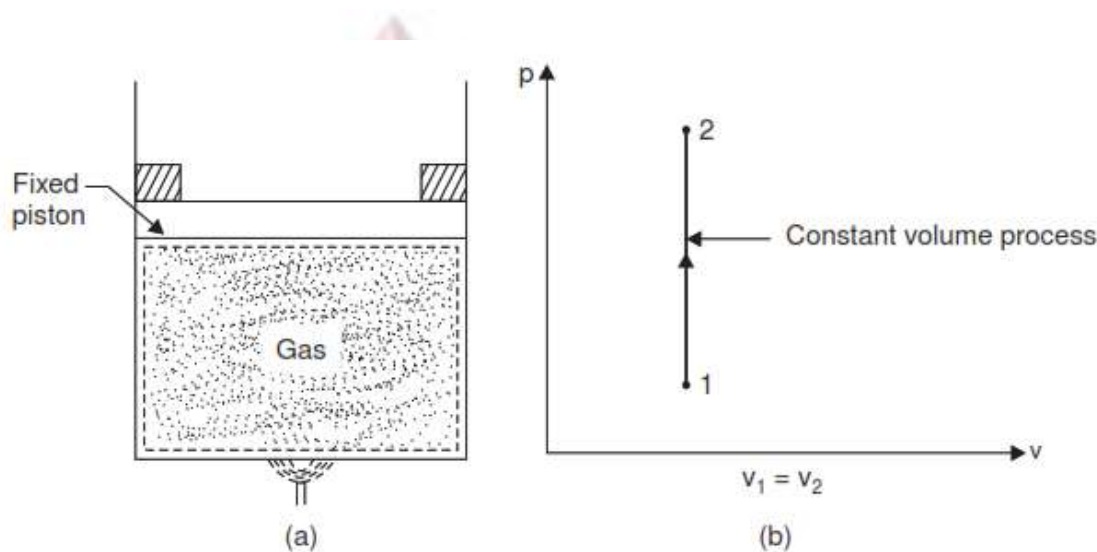


Fig. 4.5. Reversible constant volume process.

Considering mass of the working substance *unity* and applying first law of thermodynamics to the process

$$Q = (u_2 - u_1) + W \quad \dots(4.26)$$

The work done $W = \int_1^2 p dv = 0$ as $dv = 0$.

$$\therefore Q = (u_2 - u_1) = c_v(T_2 - T_1) \quad \dots[4.27 (a)]$$

where c_v = Specific heat at constant volume.

For mass, m , of working substance

$$Q = U_2 - U_1 = mc_v(T_2 - T_1) \quad \dots[4.27 (b)]$$

$$[\because mu = U]$$

2. Reversible Constant Pressure (or Isobaric) Process ($p = \text{constant}$).

It can be seen from Fig. 4.5 (b) that when the boundary of the system is inflexible as in a constant volume process, then the pressure rises when heat is supplied. Hence for a constant pressure process, the boundary must move against an external resistance as heat is supplied; for instance a gas [Fig. 4.6 (a)] in a cylinder behind a piston can be made to undergo a constant pressure process. Since the piston is pushed through a certain distance by the force exerted by the gas, then the work is done by the gas on its surroundings.

Fig. 4.6 shows the system and states before and after the heat addition at constant pressure.

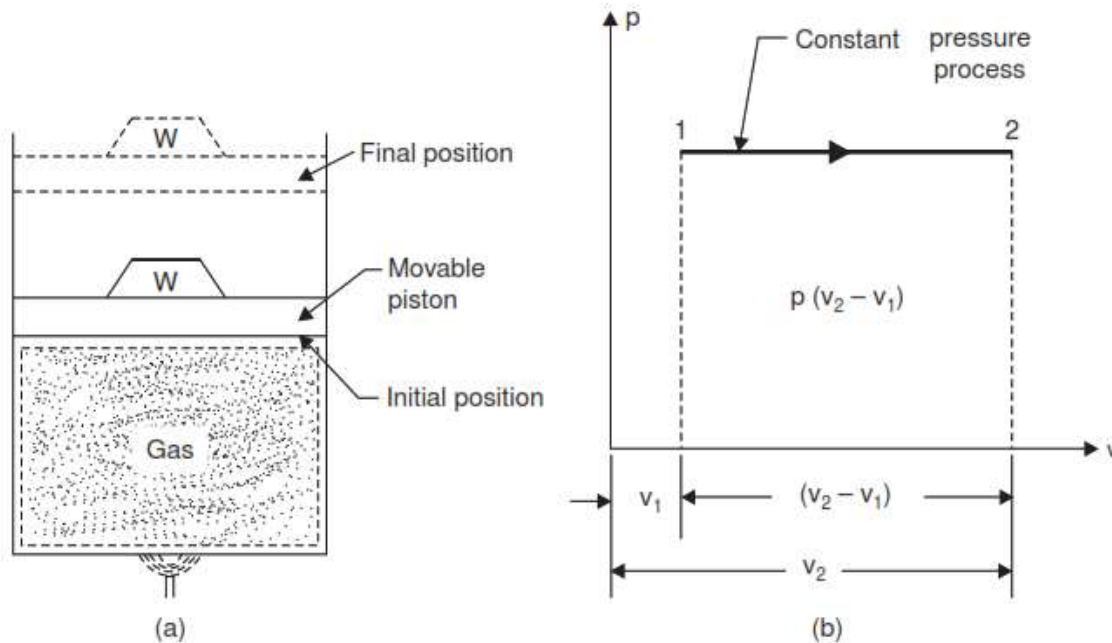


Fig. 4.6. Reversible constant pressure process.

Considering unit mass of working substance and applying first law of thermodynamics to the process

$$Q = (u_2 - u_1) + W$$

The work done, $W = \int_1^2 p dv = p(v_2 - v_1)$

$$\therefore Q = (u_2 - u_1) + p(v_2 - v_1) = u_2 - u_1 + pv_2 - pv_1$$

$$= (u_2 + pv_2) - (u_1 + pv_1) = h_2 - h_1 \quad [\because h = u + pv]$$

or $Q = h_2 - h_1 = c_p (T_2 - T_1)$... (4.28)

where h = Enthalpy (specific), and

c_p = Specific heat at constant pressure.

For mass, m , of working substance

$$Q = H_2 - H_1 = mc_p (T_2 - T_1) \quad \dots [4.28 (a)]$$

3. Reversible Temperature (or Isothermal) Process ($pv = \text{constant}$, $T = \text{constant}$) :

A process at a constant temperature is called an isothermal process. When a working substance in a cylinder behind a piston expands from a high pressure to a low pressure there is a tendency for the temperature to fall. In an isothermal expansion heat must be added continuously in order to keep the temperature at the initial value. Similarly in an isothermal compression heat must be removed from the working substance continuously during the process.

Fig. 4.7 shows the system and states before and after the heat addition at constant temperature.

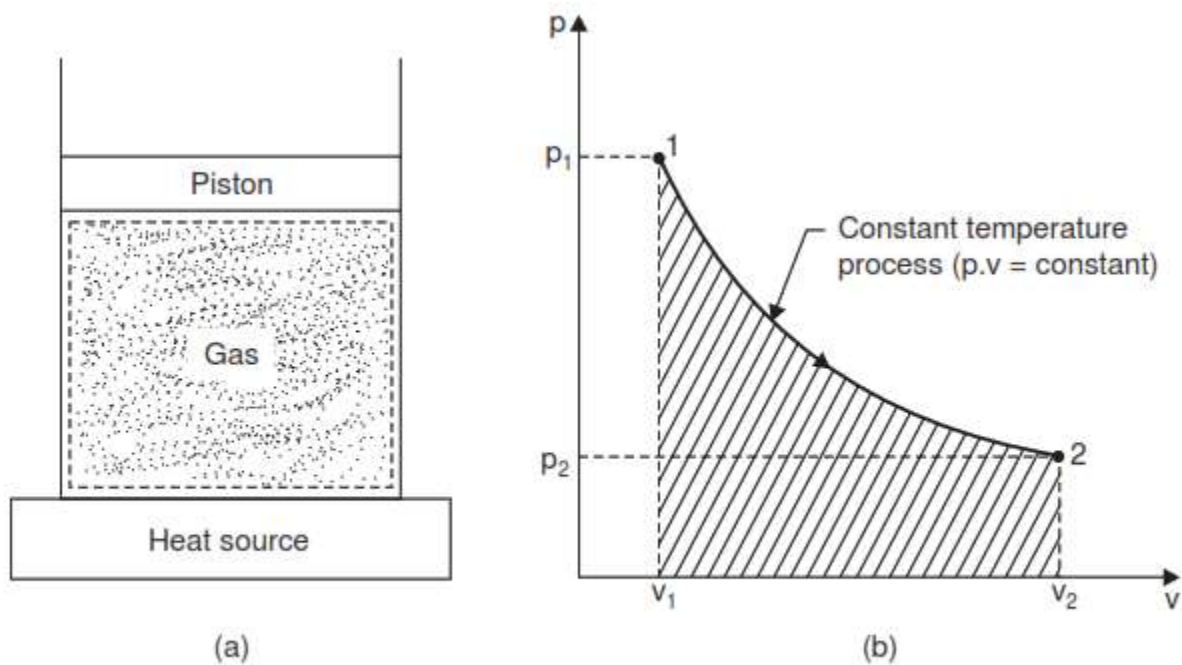


Fig. 4.7. Reversible isothermal process.

Considering unit mass of working substance and applying first law to the process

$$\begin{aligned}
 Q &= (u_2 - u_1) + W \\
 &= c_v (T_2 - T_1) + W \\
 &= 0 + W \quad [\because T_2 = T_1]
 \end{aligned}$$

The work done, $W = \int_1^2 p dv$

In this case $pv = \text{constant}$ or $p = \frac{C}{v}$ (where $C = \text{constant}$)

$$\therefore W = \int_{v_1}^{v_2} C \frac{dv}{v} = C [\log_e v]_{v_1}^{v_2} = C \log_e \frac{v_2}{v_1}$$

The constant C can either be written as $p_1 v_1$ or as $p_2 v_2$, since

$$p_1 v_1 = p_2 v_2 = \text{constant}, C$$

$$\text{i.e.,} \quad W = p_1 v_1 \log_e \frac{v_2}{v_1} \text{ per unit mass of working substance}$$

$$\text{or} \quad W = p_2 v_2 \log_e \frac{v_2}{v_1} \text{ per unit mass of working substance}$$

$$\therefore \quad Q = W = p_1 v_1 \log_e \frac{v_2}{v_1} \quad \dots(4.29)$$

For mass, m , of the working substance

$$Q = p_1 V_1 \log_e \frac{V_2}{V_1} \quad \dots[4.29 (a)]$$

$$\text{or} \quad Q = p_1 V_1 \log_e \frac{p_1}{p_2} \quad \left[\because \frac{V_2}{V_1} = \frac{p_1}{p_2} \right] \quad \dots[4.29 (b)]$$

4. Reversible Adiabatic Process $(pv)^\gamma = \text{constant}$:

An adiabatic process is one in which no heat is transferred to or from the fluid during the process. Such a process can be reversible or irreversible. The reversible adiabatic non-flow process will be considered in this section.

Considering unit mass of working substance and applying first law to the process

$$Q = (u_2 - u_1) + W$$

$$0 = (u_2 - u_1) + W$$

$$W = (u_1 - u_2) \text{ for any adiabatic process} \quad \dots (4.30)$$

Eqn. (4.30) is true for an adiabatic process whether the process is reversible or not. In an adiabatic expansion, the work done W by the fluid is at the expense of a reduction in the internal energy of the fluid. Similarly in an adiabatic compression process all the work done on the fluid goes to increase the internal energy of the fluid.

For an adiabatic process to take place, perfect thermal insulation for the system must be available.

Expression for work W :

A reversible adiabatic process for a perfect gas is shown on a $p-v$ diagram in Fig. 4.8 (b).

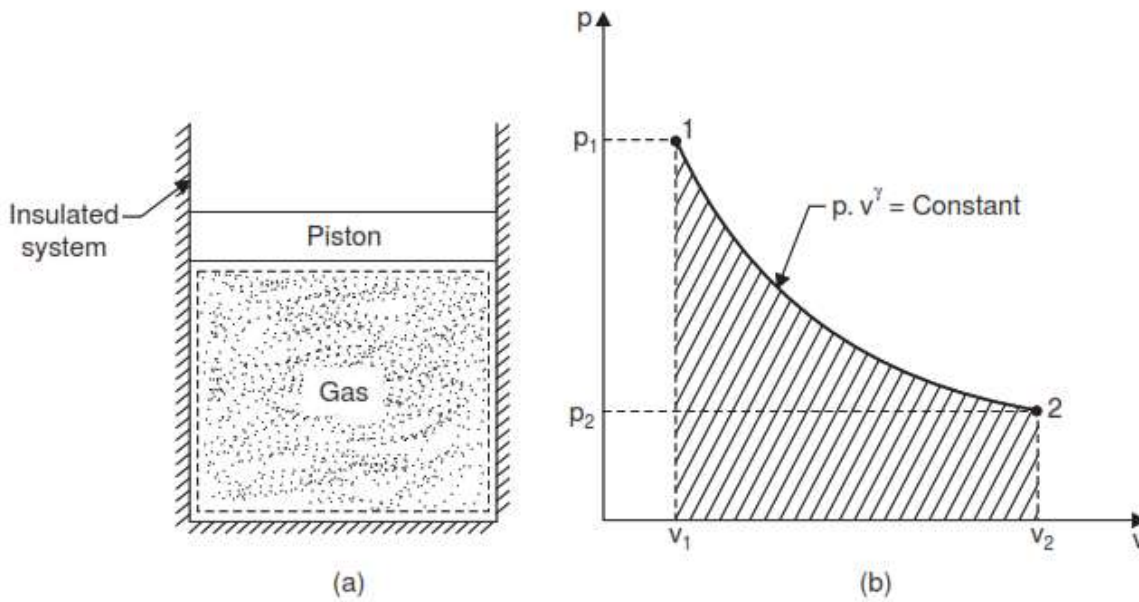


Fig. 4.8. Reversible adiabatic process.

The work done is given by the shaded area, and this area can be evaluated by integration.

$$i.e., \quad W = \int_{v_1}^{v_2} p \, dv$$

Therefore, since $pv^\gamma = \text{constant}$, C , then

$$W = \int_{v_1}^{v_2} C \frac{dv}{v^\gamma} \quad \left[\because p = \frac{C}{v^\gamma} \right]$$

$$i.e., \quad W = C \int_{v_1}^{v_2} \frac{dv}{v^\gamma} = C \left[\frac{v^{-\gamma+1}}{-\gamma+1} \right]_{v_1}^{v_2}$$

$$= C \left(\frac{v_2^{-\gamma+1} - v_1^{-\gamma+1}}{1-\gamma} \right) = C \left(\frac{v_1^{-\gamma+1} - v_2^{-\gamma+1}}{\gamma-1} \right)$$

The constant in this equation can be written as $p_1 v_1^\gamma$ or as $p_2 v_2^\gamma$. Hence,

$$W = \frac{p_1 v_1^\gamma v_1^{-\gamma+1} - p_2 v_2^\gamma v_2^{-\gamma+1}}{\gamma-1} = \frac{p_1 v_1 - p_2 v_2}{\gamma-1}$$

$$i.e., \quad W = \frac{p_1 v_1 - p_2 v_2}{\gamma-1} \quad \dots(4.32)$$

$$\text{or} \quad W = \frac{R(T_1 - T_2)}{\gamma-1} \quad \dots(4.33)$$

REFERENCES:-

- 1. R. K. Rajput, Basic Mechanical Engineering, Laxmi Publications**
- 2. B. Agrawal, Mechanical Engineering, John Wiley & Sons**



ELEMENT OF MECHANICAL ENGINEERING

LECTURE -06 (UNIT-01)

Syllabus :- Second Law of thermodynamics: Kelvin-Planck and Clausius statements

6.1 Polytropic Reversible Process ($p v^n = \text{constant}$) :

It is found that many processes in practice approximate to a reversible law of form $p v^n = \text{constant}$, where n is a constant. Both vapours and perfect gases obey this type of law closely in many non-flow processes. Such processes are internally reversible.

We know that for any reversible process,

$$W = \int p \, dv$$

For a process in $p v^n = \text{constant}$, we have

$$p = \frac{C}{v^n}, \text{ where } C \text{ is a constant}$$

$$\therefore W = C \int_{v_1}^{v_2} \frac{dv}{v^n} = C \left[\frac{v^{-n+1}}{-n+1} \right] = C \left(\frac{v_2^{-n+1} - v_1^{-n+1}}{-n+1} \right)$$

$$\text{i.e., } W = C \left(\frac{v_1^{-n+1} - v_2^{-n+1}}{n-1} \right) = \frac{p_1 v_1^n v_1^{-n+1} - p_2 v_2^n v_2^{-n+1}}{n-1}$$

(since the constant C , can be written as $p_1 v_1^n$ or as $p_2 v_2^n$)

$$\text{i.e., } \text{Work done, } W = \frac{p_1 v_1 - p_2 v_2}{n-1} \quad \dots(4.39)$$

$$\text{or } W = \frac{R(T_1 - T_2)}{n-1} \quad \dots(4.40)$$

6.2 LIMITATIONS OF FIRST LAW OF THERMODYNAMICS AND INTRODUCTION TO SECOND LAW

Further the first law of thermodynamics establishes equivalence between the quantity of heat used and the mechanical work but does not specify the conditions under which conversion of heat into work is possible, neither the direction in which heat transfer can take place. This gap has been bridged by the second law of thermodynamics.

6.3. STATEMENTS OF SECOND LAW OF THERMODYNAMICS

The second law of thermodynamics has been enunciated meticulously by Clausius, Kelvin and Planck in slightly different words although both statements are basically identical. Each statement is based on an irreversible process. The first considers transformation of heat between two thermal reservoirs while the second considers the transformation of heat into work.

6.3.1. Clausius Statement

“It is impossible for a self acting machine working in a cyclic process unaided by any external agency, to convey heat from a body at a lower temperature to a body at a higher temperature”. In other words, heat of, itself, cannot flow from a colder to a hotter body.

6.3.2. Kelvin-Planck Statement

“It is impossible to construct an engine, which while operating in a cycle produces no other effect except to extract heat from a single reservoir and do equivalent amount of work”. Although the Clausius and Kelvin-Planck statements appear to be different, they are really equivalent in the sense that a violation of either statement implies violation of other.

6.4. Equivalence of Clausius Statement to the Kelvin-Planck Statement

Refer Fig. 5.2. Consider a higher temperature reservoir T_1 and low temperature reservoir

. Fig. 5.2 shows a heat pump which requires no work and transfers an amount of Q_2 from a low

temperature to a higher temperature reservoir (in violation of the Clausius statement). Let an amount of heat Q_1 (greater than Q_2) be transferred from high temperature reservoir to heat engine which develops a net work, $W = Q_1 - Q_2$ and rejects Q_2 to the low temperature reservoir. Since there is no heat interaction with the low temperature, it can be eliminated. The combined system of the heat engine and heat pump acts then like a heat engine exchanging heat with a single reservoir, which is the violation of the Kelvin-Planck statement.

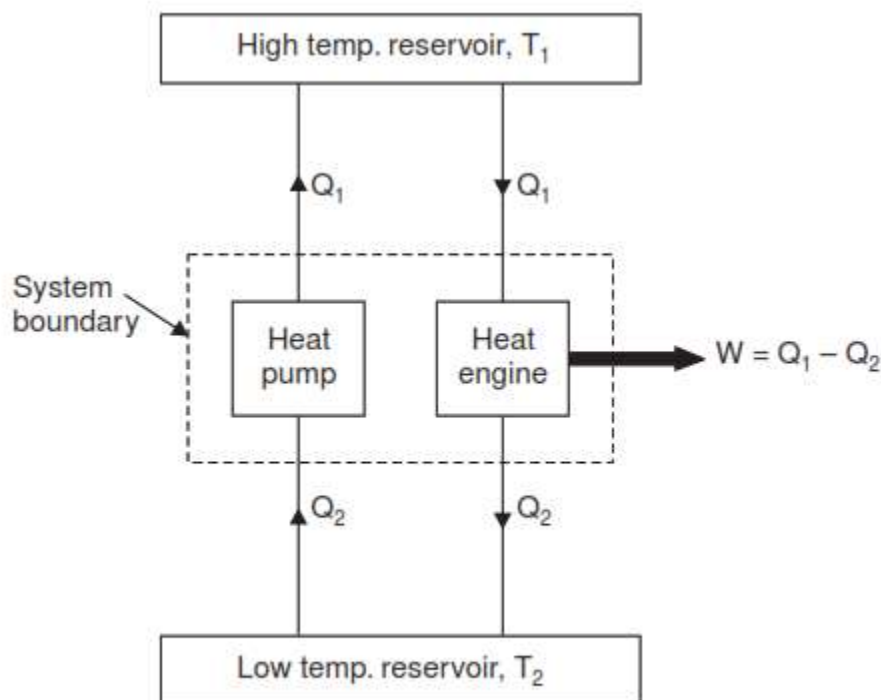


Fig. 5.2. Equivalence of Clausius statement to Kelvin-Planck statement.

REFERENCES:-

1. R. K. Rajput, Basic Mechanical Engineering, Laxmi Publications
2. B. Agrawal, Mechanical Engineering, John Wiley & Sons

ELEMENT OF MECHANICAL ENGINEERING**LECTURE -07 (UNIT-01)****Syllabus :- REVERSIBLE PROCESSES, CARNOT CYCLE,****7.1 PERFORMANCE OF HEAT ENGINES AND REVERSED HEAT ENGINES**

A heat engine is used to produce the maximum work transfer from a given positive heat transfer.

The measure of success is called the thermal efficiency of the engine and is defined by the ratio :

Thermal efficiency, $\eta_{th} = W / Q_1$(1)

where, W = Net work transfer from the engine,

and Q_1 = Heat transfer to engine.

For a reversed heat engine [Fig. 5.1 (b)] acting as a refrigerator when the purpose is to achieve the maximum heat transfer from the cold reservoir, the measure of success is called the co-efficient of performance (C.O.P.). It is defined by the ratio :

Co-efficient of performance, $(C.O.P.)_{ref.} = Q_2 / W$(5.2)

where, Q_2 = Heat transfer from cold reservoir, and

W = The net work transfer to the refrigerator.

For a reversed heat engine [Fig. 5.1 (b)] acting as a heat pump, the measure of success is again called the co-efficient of performance. It is defined by the ratio :

Co-efficient of performance, $(C.O.P.)_{heat\ pump} = Q_1 / W$(5.3)

where, Q_1 = Heat transfer to hot reservoir, and

W = Net work transfer to the heat pump.

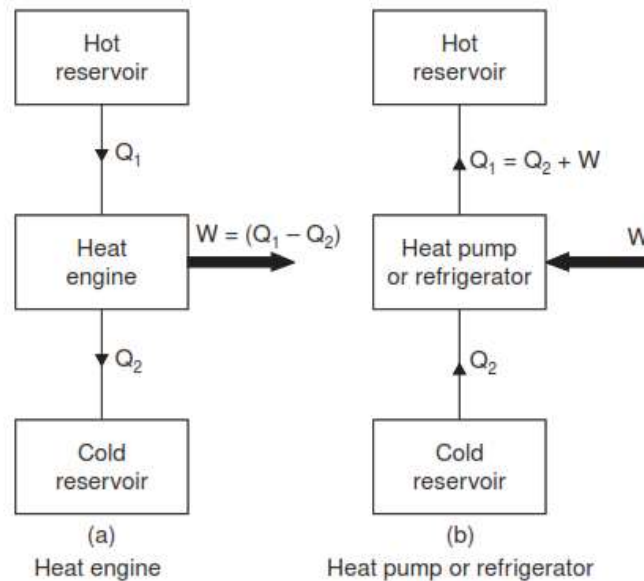


Fig. 5.1

In all the above three cases application of the first law gives the relation $Q_1 - Q_2 = W$, and this can be used to rewrite the expressions for thermal efficiency and co-efficient of performance solely in terms of the heat transfers.

$$\eta_{th} = \frac{Q_1 - Q_2}{Q_1} \quad \dots(5.4)$$

$$(C.O.P.)_{ref} = \frac{Q_2}{Q_1 - Q_2} \quad \dots(5.5)$$

$$(C.O.P.)_{heat\ pump} = \frac{Q_1}{Q_1 - Q_2} \quad \dots(5.6)$$

It may be seen that η_{th} is always less than unity and $(C.O.P.)_{heat\ pump}$ is always greater than unity.

7.2. REVERSIBLE PROCESSES

A reversible process should fulfill the following conditions :

1. The process should not involve friction of any kind.
2. Heat transfer should not take place with finite temperature difference.
3. The energy transfer as heat and work during the forward process should be identically equal to energy transfer as heat and work during the reversal of the process.
4. There should be no free or unrestricted expansion.
5. There should be no mixing of the fluids.
6. The process must proceed in a series of equilibrium states.

Some examples of ideal reversible processes are :

- (i) Frictionless adiabatic expansion or compression ;
- (ii) Frictionless isothermal expansion or compression ;
- (iii) Condensation and boiling of liquids.

Some examples of irreversible processes are :

- (i) Combustion process ;
- (ii) Mixing of two fluids ;
- (iii) All processes involving friction ;
- (iv) Flow of electric current through a resistance ;
- (v) Heat flow from a higher temperature to lower temperature.

Reversible processes are preferred because the devices which produce work such as engines and turbines, reversible process of the working fluid delivers more work than the corresponding irreversible processes. Also in case of fans, compressors, refrigerators and pumps less power input is required when reversible processes are used in place of corresponding irreversible ones.

PERPETUAL MOTION MACHINE OF THE SECOND KIND

- A machine which violates the first law of thermodynamics is called the perpetual motion machine of the first kind (PMM1). Such a machine creates its own energy from nothing and does not exist.
- Without violating the first law, a machine can be imagined which would continuously absorb heat from a single thermal reservoir and would convert this heat completely into work. The efficiency of such a machine would be 100 per cent. This machine is called the perpetual motion machine of the second kind (PMM2).

Fig. 5.3 shows the perpetual motion machine of the second kind. A machine of this kind will evidently violate the second law of thermodynamics.

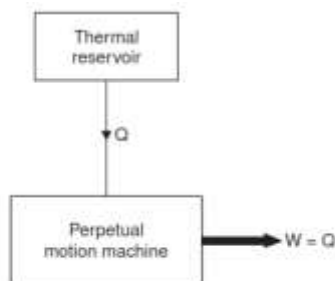
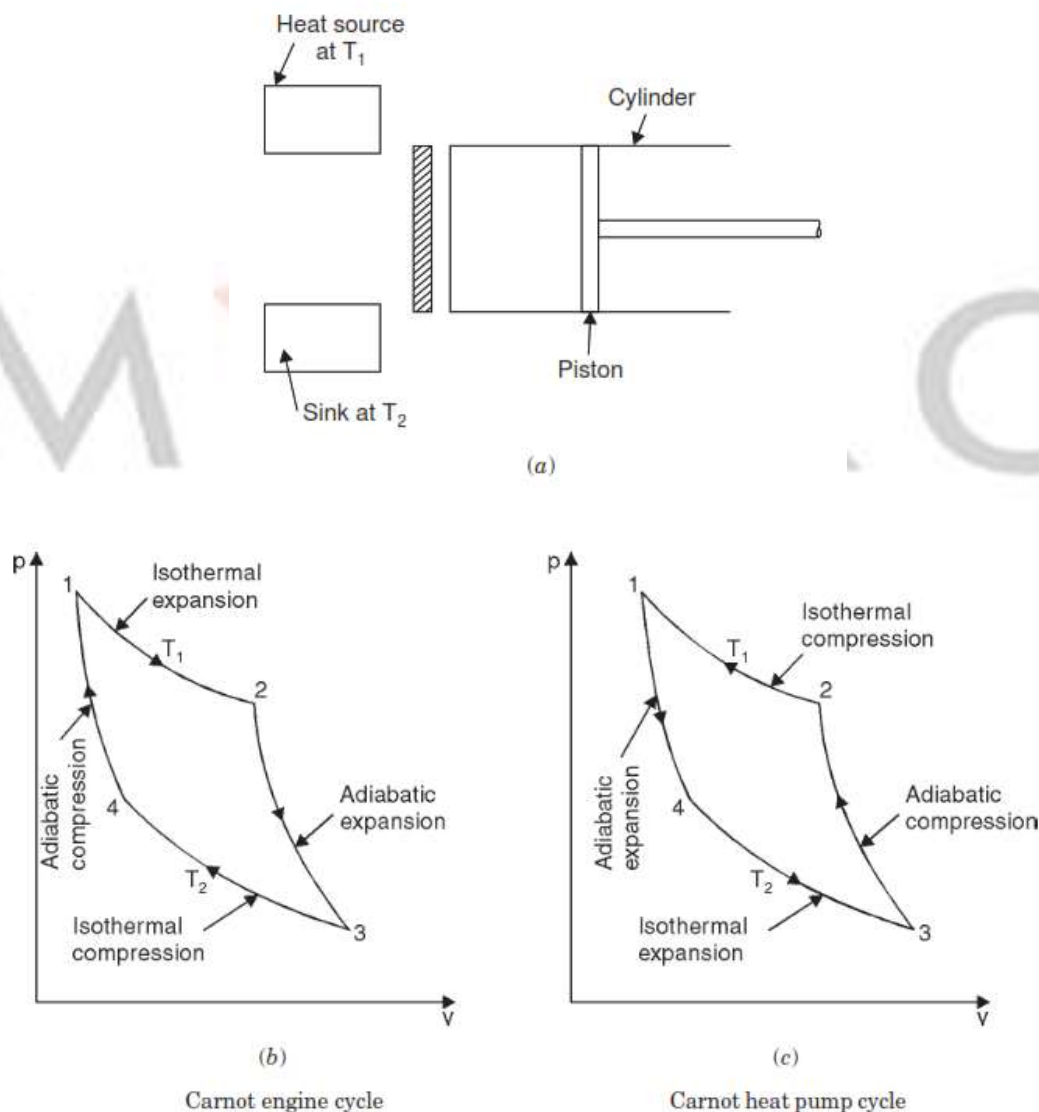


Fig. 5.3. Perpetual motion machine of second kind (PMM2).

CARNOT CYCLE

The cycle was first suggested by a French engineer Sadi Carnot in 1824 which works on reversible cycle and is known as Carnot cycle.

Any fluid may be used to operate the Carnot cycle (Fig. 5.5) which is performed in an engine cylinder the head of which is supposed alternatively to be perfect conductor or a perfect insulator of a heat. Heat is caused to flow into the cylinder by the application of high temperature energy source to the cylinder head during expansion, and to flow from the cylinder by the application of a lower temperature energy source to the head during compression.



The assumptions made for describing the working of the Carnot engine are as follows :

- (i) The piston moving in a cylinder does not develop any friction during motion.
- (ii) The walls of piston and cylinder are considered as perfect insulators of heat.
- (iii) The cylinder head is so arranged that it can be a perfect heat conductor or perfect heat insulator.
- (iv) The transfer of heat does not affect the temperature of source or sink.
- (v) Working medium is a perfect gas and has constant specific heat.
- (vi) Compression and expansion are reversible.

Following are the four stages of Carnot cycle :

Stage 1. (Process 1-2). Hot energy source is applied. Heat Q_1 is taken in whilst the fluid expands isothermally and reversibly at constant high temperature T_1 .

Stage 2. (Process 2-3). The cylinder becomes a perfect insulator so that no heat flow takes place. The fluid expands adiabatically and reversibly whilst temperature falls from T_1 to T_2 .

Stage 3. (Process 3-4). Cold energy source is applied. Heat Q_2 flows from the fluid whilst it is compressed isothermally and reversibly at constant lower temperature T_2 .

Stage 4. (Process 4-1). Cylinder head becomes a perfect insulator so that no heat flow occurs. The compression is continued adiabatically and reversibly during which temperature is raised from T_2 to T_1 .

According to first law of the thermodynamics the work obtained is equal to the difference between the heat supplied by the source (Q_1) and the heat rejected to the sink (Q_2).

$$W = Q_1 - Q_2$$

$$\text{Also, thermal efficiency, } \eta_{th} = \frac{\text{Work done}}{\text{Heat supplied by the source}} = \frac{Q_1 - Q_2}{Q_1}$$

$$= 1 - \frac{Q_2}{Q_1} \left(= 1 - \frac{T_2}{T_1} \right)$$

$$\left[\begin{array}{l} \because Q_1 = m c_p T_1 \\ Q_2 = m c_p T_2 \\ \text{where, } m = \text{mass of fluid.} \end{array} \right]$$

Such an engine since it consists entirely of reversible processes, can operate in the reverse direction so that it follows the cycle shown in Fig. 5.5 (b) and operates as a heat pump. Q_2 is being taken in at the lower temperature T_2 during the isothermal expansion (process 4-3) and heat Q_1 is being rejected at the upper temperature T_1 (process 2-1). Work W will be needed to drive the pump.

Again, the enclosed area represents this work which is exactly equal to that flowing from it when used as engine.

The Carnot cycle cannot be performed in practice because of the following reasons:

1. It is impossible to perform a frictionless process.
2. It is impossible to transfer the heat without temperature potential.
3. Isothermal process can be achieved only if the piston moves very slowly to allow heat transfer so that the temperature remains constant. Adiabatic process can be achieved only if the piston moves as fast as possible so that the heat transfer is negligible due to very short time available. The isothermal and adiabatic processes take place during the same stroke therefore the piston has to move very slowly for part of the stroke and it has to move very fast during remaining stroke. This variation of motion of the piston during the same stroke is not possible.

REFERENCES:-

1. R. K. Rajput, Basic Mechanical Engineering, Laxmi Publications
2. B. Agrawal, Mechanical Engineering, John Wiley & Sons

ELEMENT OF MECHANICAL ENGINEERING

LECTURE -08 (UNIT-01)

Syllabus :- CARNOT THEOREM, REVERSED CARNOT CYCLE

CARNOT'S THEOREM

“It states that of all engines operating between a given constant temperature source and a given constant temperature sink, none has a higher efficiency than a reversible engine”.

Refer Fig. 5.6.

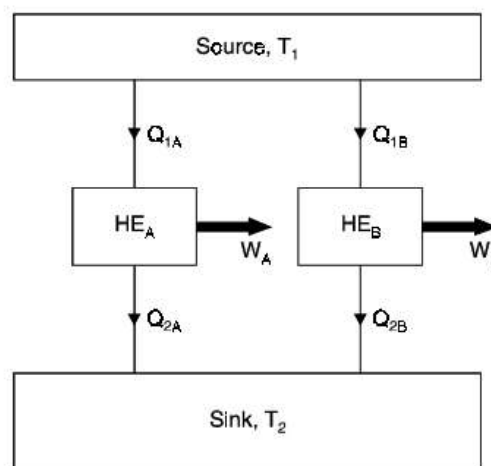


Fig. 5.6. Two cyclic heat engines HE_A and HE_B operating between the same source and sink, of which HE_B is reversible.

HE_A and HE_B are the two engines operating between the given source at temperature T_1 and the given sink at temperature T_2 .

Let HE_A be any heat engine and HE_B be any reversible heat engine. We have to prove that efficiency of HE_B is more than that of HE_A .

Let us assume that $\eta_A > \eta_B$.

Let the rates of working of the engines be such that

$$\begin{aligned} \text{Since } Q_{1A} &= Q_{1B} = Q_1 \\ \eta_A &> \eta_B \\ \frac{W_A}{Q_{1A}} &> \frac{W_B}{Q_{1B}} \\ \therefore W_A &> W_B \end{aligned}$$

Now, let HE_B be reversed. Since HE_B is a reversible heat engine, the magnitudes of heat and work transfer quantities will remain the same, but their directions will be reversed, as shown in Fig. 5.7. Since $W_A > W_B$, some part of W_A (equal to W_B) may be fed to drive the reversed heat engine. Since $Q_{1A} = Q_{1B} = Q_1$, the heat discharged by $\exists H_B$ may be supplied to HE_A .

The source may, therefore, be eliminated (Fig. 5.8). The net result is that HE_A and $\exists H_B$ together constitute a heat engine which, operating in a cycle produces net work $W_A - W_B$ while exchanging heat with a single reservoir at T_2 . This violates the Kelvin-Planck statement of the second law. Hence the assumption that $\eta_A > \eta_B$ is wrong.

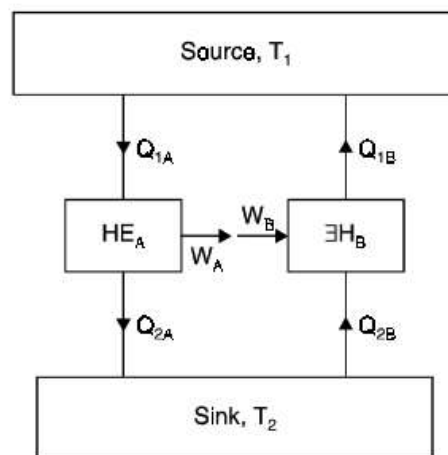


Fig. 5.7. HE_B is reversed.

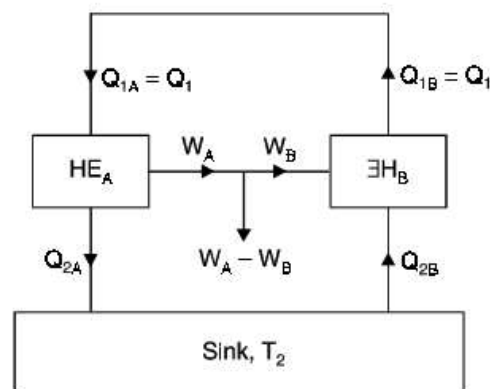


Fig. 5.8. HE_A and $\exists H_B$ together violate the Kelvin-Planck statement.

$\therefore$

$$\eta_B \geq \eta_A$$

COROLLARY OF CARNOT'S THEOREM

“The efficiency of all reversible heat engines operating between the same temperature levels is the same”.

Refer Fig. 5.6. Let both the heat engines HE_A and HE_B be reversible. Let us assume

$\eta_A > \eta_B$. Similar to the procedure outlined in the Article 5.9,

if HE_B is reversed to run say, as a heat pump using some part of the work output (W_A) of engine HE_A , we see that the combined system of heat pump HE_B and engine HE_A , becomes a PMM2. So

η_A cannot be greater than η_B . Similarly, if we assume $\eta_B > \eta_A$ and reverse the engine HE_A

we observe that η_B cannot be greater than η_A

$\therefore \eta_A = \eta_B$.

Since the efficiencies of all reversible engines operating between the same heat reservoirs are the same, the efficiency of a reversible engine is independent of the nature or amount of the working substance undergoing the cycle.

EFFICIENCY OF THE REVERSIBLE HEAT ENGINE

The efficiency of a reversible heat engine in which heat is received solely at T_1 is found to be

$$\eta_{rev.} = \eta_{max} = 1 - \left(\frac{Q_2}{Q_1} \right)_{rev.} = 1 - \frac{T_2}{T_1}$$

or

$$\eta_{rev.} = \frac{T_1 - T_2}{T_1}$$

From the above expression, it may be noted that as T_2 decreases and T_1 increases, the efficiency of the reversible cycle increases.

Since η is always less than unity, T_2 is always greater than zero and + ve.

The C.O.P. of a refrigerator is given by

$$(C.O.P.)_{ref.} = \frac{Q_2}{Q_1 - Q_2} = \frac{1}{\frac{Q_1}{Q_2} - 1}$$

For a *reversible refrigerator*, using

$$\frac{Q_1}{Q_2} = \frac{T_1}{T_2}$$

$$(C.O.P.)_{rev.} = \frac{1}{\frac{T_1}{T_2} - 1}$$

$$\therefore [(C.O.P.)_{ref.}]_{rev.} = \frac{T_2}{T_1 - T_2} \quad \dots(5.14)$$

Similarly, for a *reversible heat pump*

$$[(C.O.P.)_{heat\ pump}]_{rev.} = \frac{T_1}{T_1 - T_2} \quad \dots(5.15)$$

Example 5.1. A heat engine receives heat at the rate of 1500 kJ/min and gives an output of 8.2 kW. Determine :

(i) The thermal efficiency ;

(ii) The rate of heat rejection.

Solution. Heat received by the heat engine,

$$Q_1 = 1500 \text{ kJ/min} \\ = \frac{1500}{60} = 25 \text{ kJ/s}$$

Work output, $W = 8.2 \text{ kW} = 8.2 \text{ kJ/s}$.

(i) Thermal efficiency, $\eta_{th} = \frac{W}{Q_1}$

$$= \frac{8.2}{25} = 0.328 = 32.8\%$$

Hence, **thermal efficiency = 32.8%. (Ans.)**

(ii) Rate of heat rejection,

$$Q_2 = Q_1 - W = 25 - 8.2 \\ = 16.8 \text{ kJ/s}$$

Hence, **the rate of heat rejection = 16.8 kJ/s. (Ans.)**

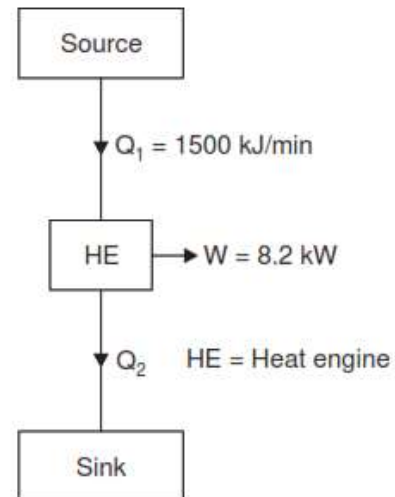


Fig. 5.9

Example 5.2. During a process a system receives 30 kJ of heat from a reservoir and does 60 kJ of work. Is it possible to reach initial state by an adiabatic process ?

Solution. Heat received by the system = 30 kJ

Work done = 60 kJ

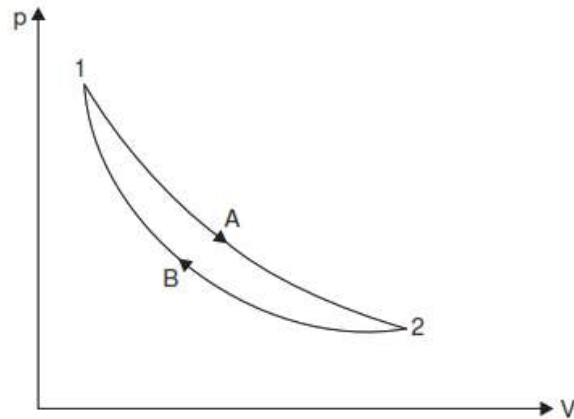


Fig. 5.10

Process 1-2 : By first law of thermodynamics,

$$Q_{1-2} = (U_2 - U_1) + W_{1-2}$$

$$30 = (U_2 - U_1) + 60 \quad \therefore \quad (U_2 - U_1) = -30 \text{ kJ.}$$

Process 2-1 : By first law of thermodynamics,

$$Q_{2-1} = (U_1 - U_2) + W_{2-1}$$

$$\therefore \quad 0 = 30 + W_{2-1} \quad \therefore \quad W_{2-1} = -30 \text{ kJ.}$$

Thus 30 kJ work has to be done *on the system* to restore it to original state, by adiabatic process.

Example 5.3. Find the co-efficient of performance and heat transfer rate in the condenser of a refrigerator in kJ/h which has a refrigeration capacity of 12000 kJ/h when power input is 0.75 kW.

Solution. Refer Fig. 5.11.

Refrigeration capacity, $Q_2 = 12000 \text{ kJ/h}$

Power input, $W = 0.75 \text{ kW} (= 0.75 \times 60 \times 60 \text{ kJ/h})$

Co-efficient of performance, C.O.P. :

Heat transfer rate :

$$(\text{C.O.P.})_{\text{refrigerator}} = \frac{\text{Heat absorbed at lower temperature}}{\text{Work input}}$$

$$\therefore \quad \text{C.O.P.} = \frac{Q_2}{W} = \frac{12000}{0.75 \times 60 \times 60} = 4.44$$

Hence **C.O.P. = 4.44. (Ans.)**

Hence transfer rate in condenser = Q_1

According to the first law

$$Q_1 = Q_2 + W = 12000 + 0.75 \times 60 \times 60 = 14700 \text{ kJ/h}$$

Hence, **heat transfer rate = 14700 kJ/h. (Ans.)**

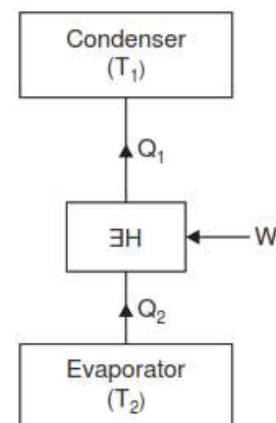


Fig. 5.11

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